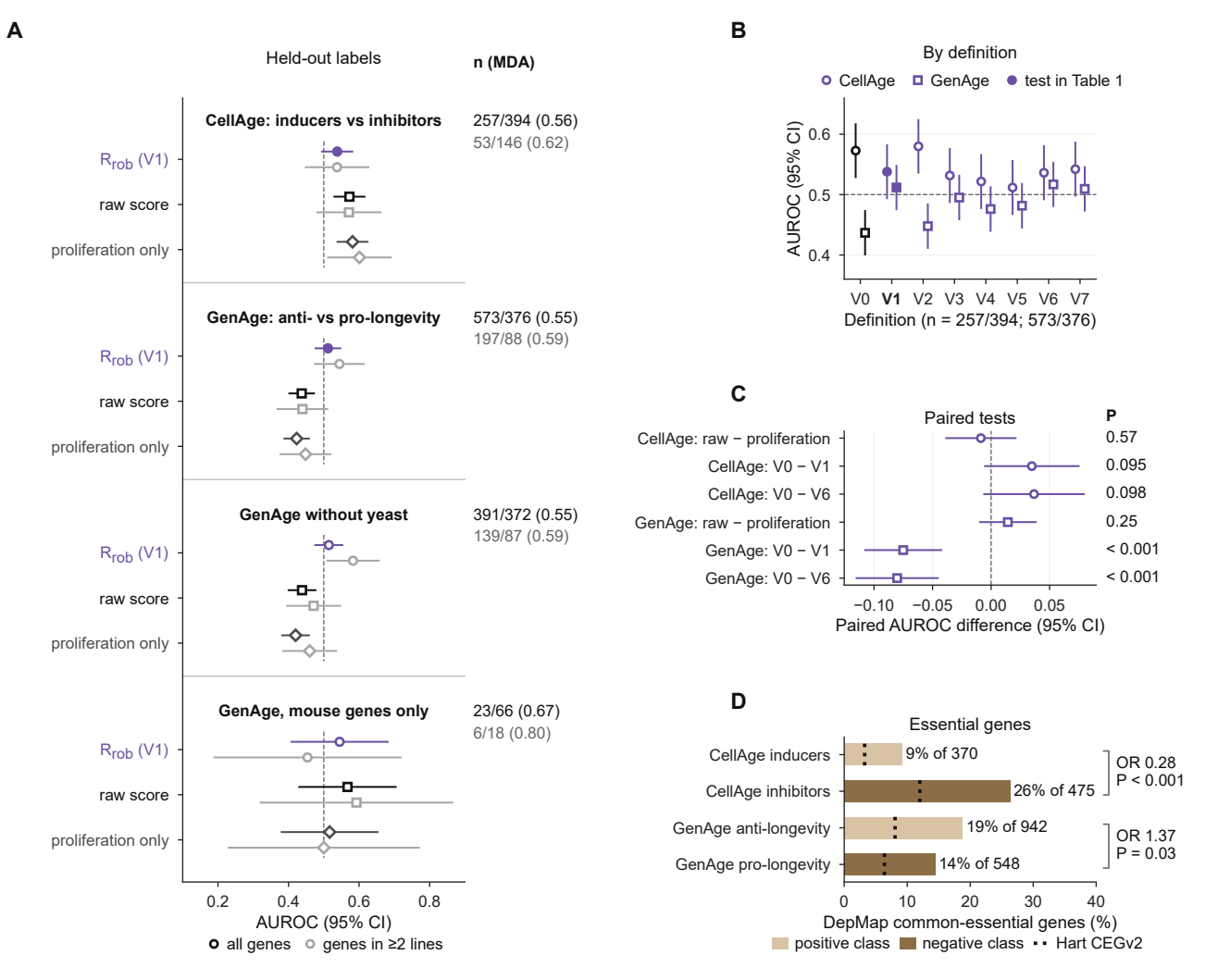

A
Held-out labels
n (MDA)
CellAge: inducers vs inhibitors
257/394 (0.56)
53/146 (0.62)
GenAge: anti- vs pro-longevity
573/376 (0.55)
197/88 (0.59)
GenAge without yeast
391/372 (0.55)
139/87 (0.59)
GenAge, mouse genes only
23/66 (0.67)
6/18 (0.80)
Rrob (V1)
raw score
proliferation only
0.2
0.4
0.6
0.8
AUROC (95% CI)
all genes
genes in ≥2 lines
B
By definition
CellAge
GenAge
test in Table 1
AUROC (95% CI)
0.6
0.5
0.4
V0
V1
V2
V3
V4
V5
V6
V7
Definition (n = 257/394; 573/376)
C
Paired tests
P
CellAge: raw − proliferation
0.57
CellAge: V0 − V1
0.095
CellAge: V0 − V6
0.098
GenAge: raw − proliferation
0.25
GenAge: V0 − V1
< 0.001
GenAge: V0 − V6
< 0.001
−0.10
−0.05
0.00
0.05
Paired AUROC difference (95% CI)
D
Essential genes
CellAge inducers
9% of 370
CellAge inhibitors
26% of 475
OR 0.28
P < 0.001
GenAge anti-longevity
19% of 942
GenAge pro-longevity
14% of 548
OR 1.37
P = 0.03
0
10
20
30
40
DepMap common-essential genes (%)
positive class
negative class
Hart CEGv2